When Does Training-Free DTW Match Learned Models on Hand Landmarks? A Data-Efficiency and Signer-Independence Study of Isolated Sign Recognition

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Abstract

We ask when matching a sign's hand landmarks against stored examples with dynamic time warping (DTW) is competitive with trained neural networks for isolated sign recognition. On PopSign ASL v1.0 (American Sign Language, smartphone video) with the dataset's signer-disjoint split, we compare DTW and Euclidean nearest neighbour, DTW prototypes, a 1D-CNN, a BiGRU and a small Transformer on the same MediaPipe features, varying the examples per sign from 1 to all and the vocabulary from 10 to 250 signs. The analysis was pre-registered, each method family chose its own preprocessing on validation signers, and intervals come from resampling the 8 test signers. Contrary to our pre-registered expectation, the learned models beat DTW at every number of examples per sign we tried: at 50 signs and 10 examples per sign the CNN scores 61.1% top-1 against 38.6% for DTW, and with all training data 85.9% against 63.4%. A second, pre-registered replication on ASL Citizen (10 to 300 signs, 11 test participants) supports both hypotheses again, and Euclidean nearest neighbour is at least as good as DTW in most cells on both datasets. Exploratory extensions (MiniRocket, soft-DTW prototypes, longer training, per-k tuning, richer features) raise some baselines, DTW to at best 71.3% with all PopSign data, but do not close the gap; an embedding trained on other signs is best with few examples per sign on both datasets. We report latency, training cost, ablations and limitations. This is an isolated-sign study of one sign language on two datasets; it is not translation and makes no claim about benefit to Deaf users.

1 Introduction

Isolated sign recognition assigns a label from a fixed vocabulary to a short video of one sign. It is a small part of sign language processing and is not translation: it ignores grammar, context and most non-manual signals. We study it because it is a clean setting in which to ask a practical methods question. When there are only a few labelled examples per sign, does matching a new video against stored examples with dynamic time warping (DTW) work as well as a trained neural network, and at what point does the network pull ahead?

A common recipe extracts hand and body landmarks with MediaPipe (Lugaresi et al., 2019; Zhang et al., 2020), turns them into per-frame features, and either trains a sequence model or compares sequences directly. Nearest-neighbour DTW needs no training and can be updated by adding examples, which is attractive for small, changing vocabularies. In general time-series classification, nearest-neighbour DTW with a tuned warping window is a strong baseline (Bagnall and Lines, 2014), and deep networks are competitive on large benchmarks (Ismail Fawaz et al., 2019). We have not found a controlled comparison of the two on signer-independent sign data that varies both the number of examples per sign and the vocabulary size, so we ran one.

What this paper does. On PopSign ASL v1.0 (Starner et al., 2023), with the dataset's signer-disjoint split, we compare DTW nearest-neighbour matching, DTW prototypes, Euclidean nearest neighbour, a 1D-CNN, a BiGRU and a small Transformer, all on the same MediaPipe features and the same queries. We vary the references per sign from 1 to all, the vocabulary from 10 to 250 signs, the feature set, the handling of missing hands and the DTW variant. The analysis was written down before the test data were scored (`PREREG.md`), and uncertainty is estimated by resampling signers.

Replication and extensions. Because the first result went against our expectation, we then ran a pre-registered replication on a second dataset,

ASL Citizen (Desai et al., 2023), with vocabularies up to 300 signs, and exploratory extensions with additional baselines, longer training, per-k tuning and descriptive analyses of possible explanations.

What it does not do. It does not propose a new method, claim state-of-the-art accuracy, or show that the systems help Deaf people. It covers one sign language (American Sign Language) and two datasets; the main one is smartphone selfie video with a mostly one-handed vocabulary. The work was done by hearing researchers without a Deaf co-author; Section 8 discusses what that limits.

Relation to earlier work by the same author. An earlier conference paper (Mankar et al., 2024) described a MediaPipe and DTW sign recogniser built on an open-source implementation (gabguerin, 2022). That paper compared DTW with an LSTM on a small custom dataset without signer-independent evaluation. The present study keeps that pipeline's feature idea as one baseline and replaces its evaluation.

2 Related work

Datasets. Isolated-sign datasets include WLASL (Li et al., 2020) (over 2000 words, over 100 signers) and MS-ASL (Joze and Koller, 2018) (1000 signs, over 200 signers); ASL Citizen (Desai et al., 2023) (2731 signs, 52 signers), collected from consenting contributors; PopSign (Starner et al., 2023), recorded on smartphones; LSA64 (Ronchetti et al., 2023) for Argentinian Sign Language; and LSFB-ISOL (Fink et al., 2021) for Belgian French Sign Language. We use PopSign because its licence permits derived data, its signers consented to a public database, and it ships a signer-disjoint split.

Landmark-based recognition. Skeleton and pose-based models include graph convolutional networks (Jiang et al., 2021) and a pose-based Transformer (Bohacek and Hruz, 2022). These are trained models evaluated on their own benchmarks; our smaller models are not meant to reproduce them, only to represent the learned side of a data-efficiency comparison.

DTW and template matching for signs. DTW (Sakoe and Chiba, 1978) has several variants: banded and approximate forms (Salvador and Chan, 2007), averaged prototypes (DBA, Petitjean et al., 2011) and the differentiable soft-DTW (Cuturi and Blondel, 2017). For signs, Arcanjo et al. (2024) apply FastDTW to angular features computed from MediaPipe landmarks, and Saggio et al. (2020) compare k-NN with DTW against a CNN on sensor-glove data. Vandendriessche et al. (2025) study one-shot recognition against a sign dictionary with learned embeddings. We did not find prior work that varies references per sign and vocabulary size together on signer-independent video data; our search was not exhaustive. The features we call F1 follow the hand-angle vectors of the open-source implementation (gabguerin, 2022) used in our earlier work (Mankar et al., 2024).

Ethics of sign language technology. Interdisciplinary and Deaf-led work argues that sign language AI has often been shaped by hearing researchers' priorities, rests on unrepresentative datasets and overstates communication benefits (Bragg et al., 2019, 2021; Desai et al., 2024). A survey of Deaf community members reports their views on automatic translation (Tran et al., 2023). We take these as constraints on how we scope and word this paper.

3 Data and methods

Data. We use the `game` category of PopSign ASL v1.0 (Starner et al., 2023): 250 signs recorded on smartphones, CC BY 4.0, with a training split of 31 signers, a validation split of 8 and a test split of 8. The code checks that no signer appears in two splits. Vocabularies of 10, 50, 100 and 250 signs are nested prefixes of a fixed random order of the signs. A vocabulary is reported only if every one of its signs was processed; others are marked as not run. Videos are downloaded, converted to landmarks and deleted.

Landmarks and features. We use MediaPipe Holistic 0.10.14 (Lugaresi et al., 2019) on frames resized to a 480 pixel short side. Coordinates are scaled by frame width and height before any angle is computed (the earlier implementation used normalised coordinates, which distorts angles on portrait frames). A sequence is trimmed to the span between the first and last frame with a detected hand. Feature set F1 is the 210 pairwise angles between the 21 hand connection vectors per hand (identical information to the earlier 21×21 angle matrix; Euclidean distances differ by a constant factor). F2 is wrist-centred, scale-normalised hand

shape. F3 combines F1 and F2, F4 adds hand location relative to the shoulders, F5 adds upper-body pose and F6 adds face-contour landmarks. The main grid uses F1. Missing hands are zero-filled in the main setting.

Methods. **DTW 1-NN** compares a query with every stored reference using DTW with Euclidean local cost and a Sakoe–Chiba band chosen on validation signers, and returns the label of the nearest reference. **Euclidean 1-NN** compares sequences resampled to 32 frames. **DBA prototype** replaces each sign's references by their DTW barycentre (length equal to the median reference length). The learned models are a 1D-CNN, a two-layer BiGRU and a three-layer Transformer encoder (462130 to 574130 parameters) trained with cross-entropy for a fixed number of gradient steps with no early stopping, so no validation labels enter training. DTW distances are computed by our own compiled implementation, which is tested against tslearn (Tavenard et al., 2020). Every method family chooses its own preprocessing on validation signers only (50 signs, 10 examples per sign, one draw), then it is fixed: how missing hands are handled (zero-fill, carry the last detection forward, or drop those frames), whether features are divided by their standard deviation over the references, and the DTW band (3, 5, 8 frames or none) for DTW; learning rate (10^{-3} or 3×10^{-4}) and epoch budget (20 or 80) for the learned models. Training takes $\min(\max(\lceil \text{epochs} \cdot N/64 \rceil, 200), 4000)$ gradient steps for N training examples, so the largest conditions get only a few epochs.

Protocol. References (or training examples) per sign $k \in \{1, 2, 5, 10, 20, \text{all}\}$ are drawn at random from the training split, nested across k, with three draw seeds (which also seed network initialisation). Test queries are up to three videos per test signer and sign, drawn once for the largest vocabulary and restricted to the signs of each smaller one, so all methods and vocabularies share them. A query on which no hand is ever detected is counted as an error. We report top-1 accuracy with 95% intervals from resampling the 8 test signers (10,000 resamples, percentile intervals; cf. Field and Welsh, 2007), and paired differences use the same resamples. With only 8 clusters the intervals are wide and we do not report p-values. Two hypotheses were fixed in advance and judged at 97.5% (§4); all other comparisons are exploratory.

Method	1	2	5	10	20	all
DTW 1-NN	17.2	22.1	31.0	38.6	45.9	63.4
Euclidean 1-NN	18.9	24.7	35.0	42.7	47.5	63.2
DBA prototype	17.2	15.2	13.4	13.1	12.7	12.4
1D-CNN	23.2	32.0	48.2	61.1	71.6	85.5
BiGRU	20.0	28.7	43.9	56.7	67.9	84.9
Transformer	23.7	30.3	45.9	57.0	66.2	85.9
BiGRU + aug.	21.9	28.5	44.5	57.9	67.7	85.1

Table 1: Top-1 accuracy (%) at a vocabulary of 50 signs on the 8 held-out test signers, by examples per sign (feature set F1, preprocessing chosen per method on validation signers). Intervals are in `results/tables/main_top1.md`.

Method	1	2	5	10	20	all
DTW 1-NN	8.1	10.6	15.3	18.8	22.8	35.8
Euclidean 1-NN	9.1	12.0	16.5	20.1	24.1	36.9
DBA prototype	8.1	6.6	6.3	6.1	5.2	4.8
1D-CNN	9.9	15.6	29.1	39.7	49.0	60.5
BiGRU	6.8	10.9	22.8	35.3	46.4	61.8
Transformer	10.1	14.2	24.2	33.3	43.6	63.5
BiGRU + aug.	–	–	–	–	–	–

Table 2: Top-1 accuracy (%) at the full vocabulary of 250 PopSign signs on the 8 held-out test signers (exploratory; settings frozen at 50 signs).

Second dataset and extensions. After the PopSign results we added a pre-registered replication on ASL Citizen (Section 5) and exploratory extensions on PopSign (Section 4.9): k-NN voting, Fisher-weighted DTW, MiniRocket, longer training, per-k tuning and descriptive analyses. Both were specified in a written addendum; the addendum's replication part was tagged before any ASL Citizen test video was scored.

4 Results

All numbers in this section are generated from the stored result files (`CLAIMS.md` lists each one). Intervals are 95% unless stated, from resampling the 8 test signers. Seeds are averaged first. "Examples per sign" k means references for the nearest-neighbour methods and training examples for the learned models; "all" uses every training video of the sign.

4.1 Main comparison

At a vocabulary of 50 signs and 10 examples per sign, the 1D-CNN reaches 61.1% top-1 (54.1–69.6), the Transformer 57.0% and the BiGRU 56.7%, against 38.6% (30.7–48.4) for DTW 1-NN and 42.7% for Euclidean 1-NN (Table 1, Figure 1). With all training videos the learned models reach

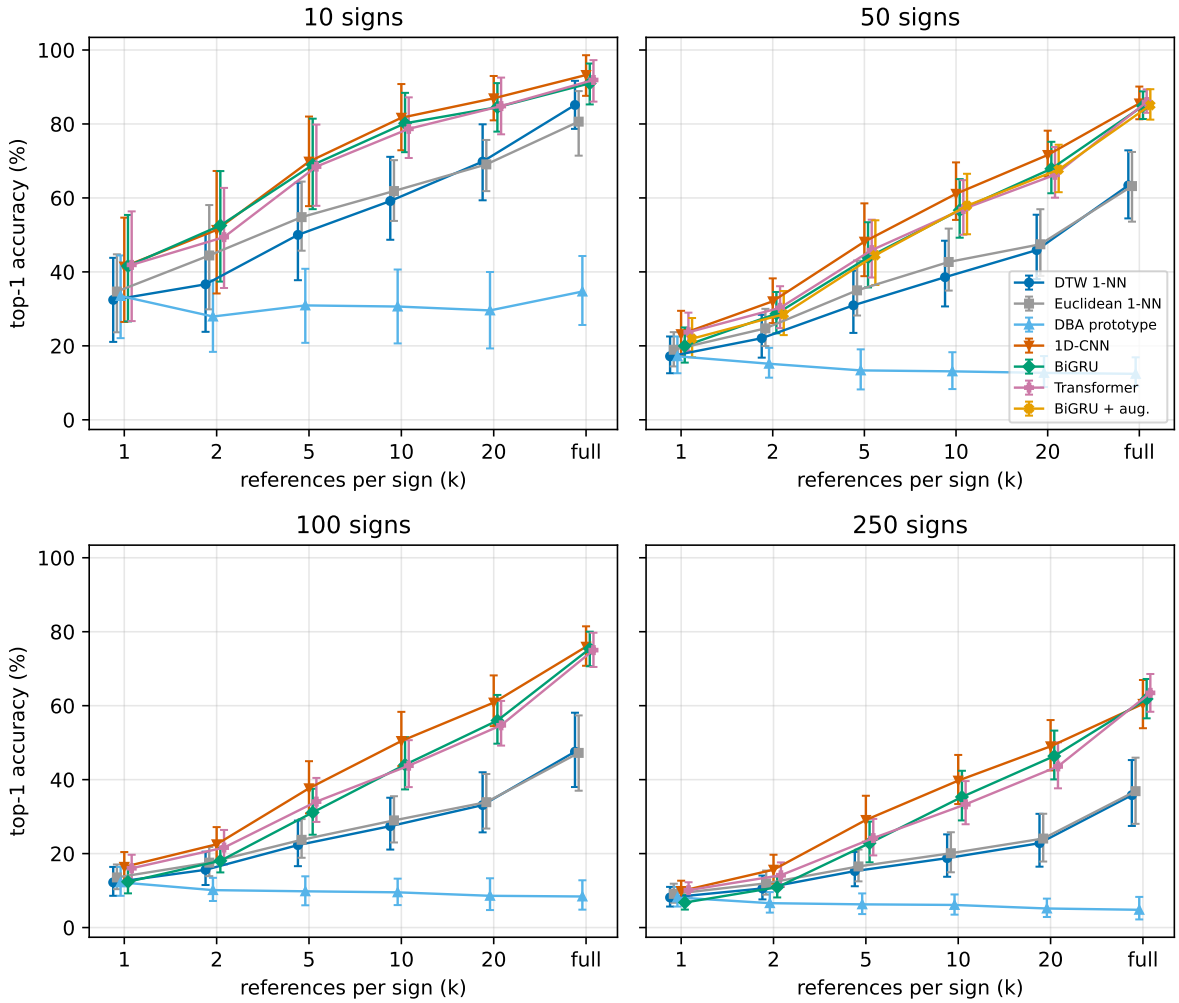


Figure 1: Top-1 accuracy versus examples per sign for vocabularies of 10 and 50 signs. Bars are 95% intervals from resampling test signers.

85.9% (Transformer), 85.5% (CNN) and 84.9% (BiGRU), and DTW 63.4%. At 10 signs the ordering is the same: 93.2% (CNN) and 85.1% (DTW) with all training videos. At 100 signs (settings frozen at 50 signs; queries drawn when that condition was built) the CNN reaches 75.9% with all training videos against 47.6% for DTW, and 16.4% against 12.2% with one example per sign. At 250 signs, the full PopSign vocabulary (same frozen settings, independent query draw), all three learned models are far above DTW with all training videos (Transformer 63.5%, BiGRU 61.8%, CNN 60.5%, against DTW 35.8% and Euclidean 36.9%), while with one example per sign every method is at or below about 10% (CNN 9.9%, BiGRU 6.8%, DTW 8.1%); the BiGRU is below DTW at $k = 1$ there (Table 2). The DBA prototype falls as examples are added at 250 signs (`results/tables/main_v250.tex`); we did not investigate why.

4.2 Pre-registered hypotheses

H1 is not supported. We expected DTW to beat each learned model at vocabulary 50 with $k \leq 5$. Instead every learned model is above DTW at $k = 1$, 2 and 5: 9 of the nine paired differences have 97.5% intervals wholly on the learned model's side of zero (`results/tables/hypotheses.csv`). For example, at $k = 1$ the CNN is 6.0 points above DTW (95% interval 4.2 to 7.7), and at $k = 5$ it is 17.2 points above (13.9 to 21.0). **H2 is supported**: with all training videos each learned model is above DTW by about 22.2 (CNN), 22.5 (Transformer) and 21.5 (BiGRU) points, and the intervals remain above zero when adjusted for three comparisons (yes).

4.3 No crossover in the tested range

We looked for the number of examples per sign at which learned models overtake DTW and found none: at every vocabulary the best learned model is at or above DTW from $k = 1$ upwards (smallest such k: 1 at 10 signs, 1 at 50, 1 at 100, 1 at 250). In 70 of the 72 comparisons of a learned model with DTW (3 models, 6 values of k, 4 vocabularies) the 95% interval excludes zero, and in 71 the learned model's point estimate is higher (exploratory; no multiplicity correction). The gap is smallest at $k = 1$, where every method is weak: at 50 signs no method exceeds 23.7% (Transformer, the best) with one example per sign. Euclidean 1-NN is not worse than DTW at 50 signs: it is ahead by 2.7, 4.0 and 4.1 points at $k = 2$, 5 and 10 (intervals exclude zero) and indistinguishable at $k = 1$, 20 and "all" (1.8, 1.6 and -0.2 points). At 10 signs DTW is clearly ahead of Euclidean 1-NN (interval excluding zero) only with all training videos (Euclidean minus DTW: -4.5 points, interval -9.0 to -1.3), while Euclidean 1-NN is ahead at $k = 2$ (7.8 points).

4.4 DTW variants

DBA prototypes (one barycentre per sign) are poor and get worse with more references per sign at 50 signs (17.2% at $k = 1$, 12.4% with all videos), so we do not recommend averaging signer variants into a single template. On one fixed draw (50 signs, 10 examples per sign, seed 0), exact DTW scores 37.2%, FastDTW (radius 1) 38.4% and the original repository's pipeline 39.1%; the intervals overlap completely (28.6–51.8% for the original pipeline), so these variants are not distinguishable at this sample size. FastDTW is not faster here: our compiled exact DTW takes 10.8 ms per query against 500 references on one thread, while the pure-Python FastDTW takes 1198.1 ms, so this comparison mostly reflects the two implementations; we did not compare the algorithms at equal implementation quality.

4.5 Preprocessing matters more for the template-matching baselines

On validation signers (50 signs, 10 examples per sign), DTW with the naive default of zero-filled missing hands and raw features scored 29.0%, against 40.4% for the configuration chosen on validation (frames without a hand dropped, standardised features, band of 8 frames). The learned models were much less sensitive: best validation accuracy with zero-filled gaps versus the chosen setting was 64.4% vs 66.6% for the CNN. Every family chose to drop frames without a detected hand. We give each family its own validation-selected preprocessing so that no method is held back by an arbitrary default; the naive default would have made the DTW baseline look 11.4 points worse on validation.

Setting (features/missing/hand slots)	$k = 5$ DTW	$k = 5$ CNN	$k =$ all DTW	$k =$ all CNN
F1/zero/slots	23.9	45.6	57.0	85.5
F1/ffill/slots	26.9	43.7	59.5	82.7
F1/drop/slots	31.0	48.2	63.4	85.5
F1/zero/dominant	28.0	59.8	54.3	88.6
F2/zero/slots	18.0	40.4	48.5	89.7
F3/zero/slots	27.2	53.2	66.7	90.3
F4/zero/slots	27.4	53.1	67.2	93.3
F5/zero/slots	27.6	52.3	67.1	92.9
F6/zero/slots	12.4	35.4	39.8	92.6
F3/drop/slots†	36.1	55.0	70.6	89.8
F4/drop/slots†	36.3	55.4	71.3	92.6
F5/drop/slots†	34.7	52.5	70.9	93.3
F4/drop/dominant†	38.1	66.0	67.1	93.4

Table 3: Top-1 accuracy (%) at 50 signs by feature set, missing-hand handling and hand slots, for DTW 1-NN and the CNN. Rows without a dagger were pre-registered and run with the stated settings; rows marked † are an exploratory supplement added after seeing the main results (see `PREREG.md`). The F1/drop row is the main setting. Other methods and intervals: `results/tables/ablation.md`.

4.6 Feature sets, missing hands and handedness (ablations)

Table 3 shows the effect of each factor (the pre-registered feature ablations used zero-filled missing hands, so compare rows within the table). Dropping frames without a hand helps DTW (23.9% with zero-fill to 31.0% at $k = 5$) more than the CNN (45.6% to 48.2%). Putting the more active hand in the first slot (dominant-hand canonicalisation) raises the CNN at $k = 5$ from 45.6% to 59.8% and DTW from 23.9% to 28.0%, but with all training videos DTW goes down (57.0% to 54.3%). Adding hand location (F4) raises DTW with all videos from 57.0% (F1) to 67.2% and the CNN from 85.5% to 93.3%; adding upper-body pose changes little; adding face contours lowers DTW sharply (39.8% with all videos) while the CNN stays near 92.6%.

Across every setting we ran, the best DTW result was 38.1% at $k = 5$ and 71.3% with all videos, against 66.0% and 93.4% for the best CNN setting, so no feature or preprocessing choice we tried closes the gap. These maxima are over many

settings on the test signers and are therefore optimistic for both families.

4.7 Augmentation, one- and two-handed signs, and per-sign errors

Light augmentation (time stretch, noise, scale) changed the learned models by at most a few points: for the CNN at 50 signs, 25.0% versus 23.2% at $k = 1$ and 85.8% versus 85.5% with all videos. The learned models' lead over DTW is therefore not an artefact of leaving them unaugmented. We could not compare one- and two-handed signs: under the pre-registered rule (both hands detected in at least half of the hand-present frames), 0 of the 50 signs are two-handed and 50 are one-handed, consistent with PopSign's one-handed focus.

With all training videos, DTW is below 50% on 14 of the 50 signs, with the lowest being sleepy (0.0%), whereas the CNN is below 50% on 0 signs (lowest: police, 50.0%). The most frequent DTW confusion is dryer recognised as dry (10 test queries). PopSign has no phonological annotation, so we make no claim about why signs are confused; the full lists are in `results/tables`.

4.8 Cost: training and inference

Template matching needs no training but its inference cost grows with the number of references: with 500 references (10 per sign) one query takes 10.8 ms (95th percentile 34.9 ms) on one thread of an Apple M4 Max with our compiled DTW, and 547.6 ms with all 21876 training videos as references (Euclidean nearest neighbour: 2.0 and 91.0 ms). A forward pass of a trained model takes 0.13 ms (CNN), 0.25 ms (Transformer) and 0.93 ms (BiGRU) on one CPU thread, independent of the number of training examples. Training the CNN and Transformer took on average 3 and 4 seconds at 10 examples per sign and 24 and 26 seconds with all videos on the Apple GPU (BiGRU: 109 and 283 seconds; training times varied with load on the shared machine). Latency was measured with other processes running, and no low-power device was available. These costs exclude MediaPipe landmark extraction (8.8 wall-clock hours for the 50 signs on this machine).

4.9 Exploratory extension: more baselines, longer training, per-k tuning

These analyses were added after the main PopSign results and the audit that followed (`PREREG_ADDENDUM.md`, Part B); they are exploratory.

Method	1	2	5	10	20	all
DTW 1-NN (main)	17.2	22.1	31.0	38.6	45.9	63.4
Euclidean 1-NN (main)	18.9	24.7	35.0	42.7	47.5	63.2
DTW k-NN (kappa on validation)	17.2	22.1	31.0	38.6	45.9	63.4
Fisher-weighted DTW	17.2	22.4	30.3	36.6	43.5	59.2
MiniRocket + ridge	11.6	12.5	30.0	40.0	50.7	70.8
DTW, tuned per k	17.2	22.1	31.0	38.6	45.9	63.4
Soft-DTW prototype	19.7	18.7	15.7	14.4	–	–
Embedding (trained on other signs) + 1-NN	47.8	54.5	62.0	64.9	68.1	75.3
1D-CNN (main)	23.2	32.0	48.2	61.1	71.6	85.5
1D-CNN, tuned per k	23.2	32.0	48.3	61.1	71.6	85.1
1D-CNN, 200 epochs	23.2	32.1	49.1	61.9	72.8	87.9
BiGRU (main)	20.0	28.7	43.9	56.7	67.9	84.9
BiGRU, 200 epochs	20.0	30.1	45.1	58.3	67.5	86.6
Transformer (main)	23.7	30.3	45.9	57.0	66.2	85.9
Transformer, 200 epochs	23.7	30.1	45.2	57.2	66.8	86.5

Table 4: PopSign, 50 signs: top-1 accuracy (%) of additional methods and variants (exploratory) beside the main ones.

Other nearest-neighbour variants. Voting among several nearest references did not help: the number of neighbours chosen on validation was one for both DTW and Euclidean. Weighting feature dimensions by a Fisher score estimated from the references did not help either (59.2% with all examples against 63.4% for DTW).

Soft-DTW prototypes. Like the DBA prototypes, soft-DTW prototypes do not improve with more references (19.7% at $k = 1$, 14.4% at $k = 10$; run only for $k \leq 10$ at 10 and 50 signs).

A learned embedding from other signs. A CNN trained only on the other 50 signs of the 100-sign vocabulary (signs 51 to 100, the same training signers), used as a fixed embedding for nearest-neighbour search on the first 50 signs, is the best method with few examples: 47.8% with one example per sign against 23.7% for the best model trained from scratch and 17.2% for DTW, 62.0% against 48.2% (CNN) at five examples and 64.9% against 61.1% at ten. From twenty examples the from-scratch models overtake it (68.1% against 71.6%, and 75.3% against 85.9% with all examples). It needs a training corpus of other signs, which template matching does not.

A stronger time-series baseline. MiniRocket with a ridge classifier (Dempster et al., 2020) is poor with one or two examples per sign (11.6% and 12.5%, against 17.2% and 22.1% for DTW), about level with DTW at $k = 5$ and 10 (30.0% and 40.0% against 31.0% and 38.6%), and above it from $k = 20$ (50.7% against 45.9%, and 70.8% against 63.4% with all examples) but well below the neural models (85.5% for the CNN).

Tuning per k and longer training. Re-selecting every method's settings separately for $k \in \{1, 5, 10, \text{all}\}$ on validation changed no cell by more than 1.6 points. Training the learned models for 200 epochs with a step cap of 12,000 instead of the main budget changed accuracy by between 0.0 and 2.3 points for the CNN, -0.4 and 1.7 for the BiGRU and -0.7 and 0.6 for the Transformer across all k (Table 4), so the main results were not far from converged and the gap to DTW did not close.

4.10 Descriptive analyses of why (exploratory)

We looked at three possible explanations; none clearly accounts for the gap. (1) *Hand slot.* MediaPipe assigns each hand to a left or right slot, and test signers differ: each signer uses predominantly one slot (at least 93.6% of hand-present frames), with the share showing only the left slot between 0.1% and 98.7% and only the right slot between 0.3% and 99.3% across signers; 3 of 8 mostly use the left slot. Swapping the slots of the test queries lowers accuracy only modestly (DTW with all examples 63.4% to 59.2%, CNN 85.6% to 83.6%), so the models are not tied to a single slot. The large gain from putting the more active hand in the first slot (Table 3) is consistent with pooling a sign's few examples across signers who use different hands, which we did not test directly. (2) *Movement.* Per-sign wrist path length has no clear relation to the per-sign accuracy gap between the CNN and DTW (Spearman $\rho = 0.13$, 95% bootstrap interval -0.19 to 0.42 over 50 signs). (3) *Feature scale or preprocessing.* Both were tuned for DTW on validation (Section 4). These analyses are descriptive; a causal account remains open.

4.11 Coverage

On PopSign the 250-sign condition was run last, after the 10-, 50- and 100-sign results were known, with settings frozen at 50 signs; it is exploratory in that sense, and face landmarks exist only for the first 50 signs. One validation archive ("aunt") was damaged on the server and partly recovered (`PREREG_ADDENDUM.md`). The vocabulary sweep up to 300 signs is on ASL Citizen (Section 5). Of the 1038 test queries at 50 signs, 1 had no detected hand and is counted as an error; the training and validation splits have 17 and 3 such videos.

Method	1	2	5	10	all
DTW 1-NN	32.0	38.9	48.5	55.7	59.7
Euclidean 1-NN	37.8	45.7	54.2	61.2	66.2
DBA prototype	32.0	37.1	43.4	49.1	49.8
1D-CNN	38.2	51.8	68.5	76.0	80.2
BiGRU	39.9	52.8	67.3	74.4	78.5
Transformer	42.9	55.3	69.8	77.0	81.6

Table 5: ASL Citizen: top-1 accuracy (%) at 50 signs on the 11 held-out test participants, by examples per sign. Intervals are in `results_aslc/tables/main_top1.md`.

5 Pre-registered replication on ASL Citizen

After the PopSign results were known we wrote a pre-registration addendum (`PREREG_ADDENDUM.md`), tagged in git before any ASL Citizen test video was scored, that fixes this replication: data, vocabularies, methods, tuning and two hypotheses.

Data and design. ASL Citizen (Desai et al., 2023) has official splits by participant (35 training, 6 validation and 11 test participants, disjoint). Its licence allows non-commercial research use but forbids distributing the data or modifications of it, so we fetched only the videos needed, extracted hand and pose landmarks with the same MediaPipe configuration, deleted the videos, extracted no face landmarks, and commit only aggregate results. Vocabularies are nested prefixes of a fixed random order of the 2,731 signs: 10, 50, 100 and 300 signs, with up to $k = 10$ or all training examples per sign (about 15 per sign; $k = 20$ is not available). At 300 signs there are 4427 training videos and 3702 test queries. The methods and the tuning procedure are those of Section 3, with settings chosen on the validation participants only. The validation-chosen preprocessing differed from PopSign: 4 of 5 families chose to carry the last hand detection forward and 1 chose to drop frames without a hand. The naive default for DTW (zero-fill, raw features, unconstrained) scored 58.0% on validation against 70.2% for the chosen configuration.

Pre-registered hypotheses. H3: at 50 signs and $k \in \{1, 2, 5\}$ each learned model is above DTW. H4: at 50 signs with all training videos at least one learned model is above DTW. **Both are supported.** 9 of the 9 H3 comparisons have 97.5% intervals wholly on the learned model's side of zero

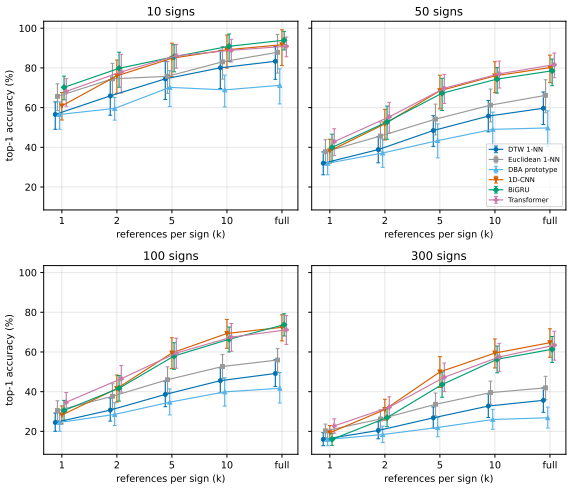


Figure 2: ASL Citizen: top-1 accuracy versus examples per sign for vocabularies of 10, 50, 100 and 300 signs. Bars are 95% intervals from resampling test participants.

(the CNN is 6.2 points above DTW at $k = 1$, 95% interval 5.0 to 7.5, and 20.1 points at $k = 5$), and H4 holds for each model; the check adjusted for three models (99.17% intervals) also holds: yes. So the main PopSign finding replicates on a second dataset collected from different signers with different devices.

Vocabulary size. Accuracy falls for every method as the vocabulary grows from 10 to 300 signs, and the ordering stays the same (Figure 2). With all training videos at 300 signs the CNN reaches 64.7% (57.3–71.7), the Transformer 63.4% and DTW 35.6%, a CNN-minus-DTW gap of 29.0 points (at 10 signs: 8.3 points). With one example per sign and 300 signs no method exceeds 22.7% (Transformer).

Euclidean nearest neighbour versus DTW. Euclidean 1-NN is above DTW in 20 of the 20 (vocabulary, k) cells, with the 95% interval above zero in 17 and below zero in 0 of them. On this dataset the extra flexibility of DTW's alignment did not help, as on PopSign (where Euclidean was above DTW in 20 of 24 cells and clearly above in 9).

The same extensions (exploratory). The extra baselines of Section 4.9 give the same picture on ASL Citizen (50 signs; Table 6). Neighbour voting again chose one neighbour. Fisher-weighted DTW is slightly better than plain DTW (61.1% against 59.7% with all examples) and still below Euclidean 1-NN (66.2%). MiniRocket is below

Method	1	2	5	10	all
DTW 1-NN (main)	32.0	38.9	48.5	55.7	59.7
Euclidean 1-NN (main)	37.8	45.7	54.2	61.2	66.2
DTW k-NN (kappa on validation)	32.0	38.9	48.5	55.7	59.7
Fisher-weighted DTW	33.9	40.8	51.3	57.6	61.1
MiniRocket + ridge	25.1	39.8	56.9	66.5	70.7
DTW, tuned per k	34.2	42.0	48.8	55.7	61.0
Soft-DTW prototype	36.2	41.5	49.0	54.6	–
Embedding (trained on other signs) + 1-NN	51.8	62.2	69.2	73.2	73.8
1D-CNN (main)	38.2	51.8	68.5	76.0	80.2
1D-CNN, tuned per k	41.3	54.7	70.5	76.0	82.1
1D-CNN, 200 epochs	38.2	51.8	69.1	77.2	80.7
BiGRU (main)	39.9	52.8	67.3	74.4	78.5
BiGRU, 200 epochs	39.9	53.7	68.4	75.1	78.5
Transformer (main)	42.9	55.3	69.8	77.0	81.6
Transformer, 200 epochs	42.9	55.7	70.2	77.6	81.1

Table 6: ASL Citizen, 50 signs: top-1 accuracy (%) of additional methods and variants (exploratory) beside the main ones.

DTW with one example (25.1% against 32.0%), level at two, and above it from five (56.9% against 48.5%, a difference of 8.5 points, interval 6.0 to 11.0) up to 70.7% with all examples, but stays below the neural models (81.6% for the Transformer). Selecting settings per k moves cells by up to 3.1 points and training for 200 epochs by between -0.1 and 1.2 points for the CNN (BiGRU 0.0 to 1.1, Transformer -0.5 to 0.6), neither closing the gap.

Soft-DTW prototypes and a learned embedding (exploratory). Soft-DTW prototypes do no better than plain DTW (54.6% against 55.7% at $k = 10$). A CNN trained only on 200 *other* signs and then used as a fixed embedding for nearest-neighbour search, with no training on the evaluated signs, is the best method with the fewest examples: 51.8% with one example per sign against 42.9% for the best model trained from scratch (Transformer) and 32.0% for DTW, and 62.2% against 55.3% at two examples. It is level with the from-scratch models at five (69.2%) and falls behind from about ten examples (73.8% with all examples against 81.6%), as it never adapts to the evaluated signs. This needs a training set of other signs, which template matching does not.

6 Discussion

What the data say. On both datasets, with signer-independent splits and hand-angle features, small learned models beat nearest-neighbour DTW from one example per sign upward. On PopSign the two pre-registered expectations came out as: H1 (DTW ahead with few examples) not supported, H2 (learned models ahead with all data) supported; the pre-registered replication on ASL Citizen supported both hypotheses at vocabularies

up to 300 signs. With one example per sign and 50 signs every method is weak on PopSign (the best point estimate is 23.7%). Euclidean nearest neighbour was at least as good as DTW in most cells on both datasets, and DBA prototypes were poor. Preprocessing mattered a great deal for the template-matching baselines (11.4 points on PopSign validation between the naive and the chosen setting) and little for the learned models, which is a reason to tune baselines before concluding anything about them.

Other baselines. A stronger time-series classifier (MiniRocket) is weaker than DTW with one or two examples on PopSign and with one on ASL Citizen, and better from about twenty examples on PopSign and from five on ASL Citizen, but stays far below the neural models; neighbour voting did not help and Fisher-weighted DTW helped only a little on ASL Citizen; training the neural models longer and tuning per k changed little (a few points at most). These checks make it less likely that the result comes from an under-tuned neural model or an unlucky DTW configuration, although they cannot rule out better template-matching or metric-learning methods that we did not try.

Transfer. On both datasets an embedding trained only on other signs gave the best accuracy with the fewest examples per sign (Sections 4.9 and 5; on ASL Citizen up to about five examples, on PopSign up to about ten), which weakens the case for template matching further in the few-example regime, at the price of needing a training set of other signs.

Why DTW trails is only partly examined. The descriptive analyses did not point to a clear cause: signers differ in which hand they sign with (which matters for the few-example regime, since the dominant-hand ablation lifts the neural models a lot), but swapping hand slots at test time costs little, and per-sign movement does not track the gap. Other plausible explanations remain untested: the per-frame angle features are noisy, so extra alignment freedom may bring unrelated signs closer; the learned models can weight informative dimensions while DTW weights all equally; and signer variation, which a model trained on 31 signers can absorb, is a direct cost for template matching. We therefore make no causal claim about the mechanism.

Relation to other results. In general time-series classification, tuned nearest-neighbour DTW is a strong baseline (Bagnall and Lines, 2014). Our result does not contradict that finding, because the setting differs: 420-dimensional noisy features from landmark detection, 50 to 300 classes, and test signers unseen in training; we did not investigate how far each of these differences matters. The earlier conference paper from the same author (Mankar et al., 2024) reported DTW ahead of an LSTM on a small custom dataset without a signer-independent split; we cannot re-run that experiment, and the present signer-independent results on two datasets do not support carrying its conclusion over to these settings.

Practical reading, with limits. For these datasets and features, a small learned model is the better default once a few labelled examples per sign and seconds to a few minutes of training are available (Section 4), and it is also much cheaper at inference when many reference videos exist. Template matching keeps its practical advantages of requiring no training and of letting a new sign be added by storing one example, but in our experiments that convenience cost accuracy. This is a statement about two datasets, one language and our feature choices (Section 7); it says nothing about whether any of these systems is useful to Deaf people, and accuracy on isolated signs is not communication access (Section 8).

7 Limitations

- **One sign language, two datasets.** All results are for ASL on PopSign (47 signers, selfie phone video, mostly one-handed signs) and ASL Citizen (crowd-sourced webcam video). We do not test other languages, cameras or continuous signing.
- **Few signers.** The PopSign test split has 8 signers and ASL Citizen 11 participants, so signer-clustered intervals are wide and small differences are not resolvable. Training noise (3 seeds) is smaller than signer variation.
- **Vocabulary range.** The vocabulary sweep reaches 300 signs only on ASL Citizen; on PopSign it reaches the full 250 signs (the 100- and 250-sign conditions use settings tuned at 50 signs and an independent query draw).
- **Small models, limited tuning.** The learned

models are small and tuned on a small grid at one condition (50 signs, 10 examples per sign). All three chose the longest epoch budget in the grid. Longer training and per-k tuning changed little (Section 4.9), but larger models or pre-training may change the picture.

- **Only F1 in the main grid, and a handicap risk for DTW.** The main comparison uses the hand-angle features of the earlier implementation, which carry no hand location or orientation. DTW and Euclidean matching each chose among the preprocessing options listed in Section 3 on validation signers, and richer features helped both families (Table 3), but other options, local costs or learned metrics might favour template matching more.
- **Baselines not run.** We ran one learned-embedding baseline (trained on other signs; cf. Vandendriessche et al., 2025) but not the full set of time-series classifiers in the bake-off (Bagnall et al., 2016), nor large pre-trained or end-to-end video models.
- **Landmark quality.** Results depend on MediaPipe 0.10.14 and on the resize used; the fraction of videos or frames without detected hands is reported. Face and other non-manual signals are mostly not modelled.
- **Not translation, not an application.** Top-1 accuracy on isolated signs is not communication access. We did not evaluate any system with Deaf users.
- **No phonological analysis.** Neither dataset has phonological annotation; the descriptive analyses make no claim about phonological similarity.
- **Timing.** Latency was measured on one Apple M4 Max laptop with other processes running; no low-power device was available, so embedded performance is not measured.
- **Pre-registration scope.** The PopSign plan and the ASL Citizen replication were written by the author before the respective test scoring, but are not registered reports and were not independently time-stamped; the git history is the record (and it was rewritten once to remove cache files, see `PREREG.md`). The extensions are post-hoc.
- **ASL Citizen reproducibility.** Its licence forbids redistributing the data or derived landmarks, so the replication can be re-run only by fetching the data again; we release aggregate results only.

8 Ethics and broader impact

Who did the work. This study was designed and run by hearing researchers without a Deaf co-author. Deaf-led analysis of sign language AI argues that such work often centres hearing researchers' priorities and a "communication barrier" framing (Desai et al., 2024). We chose a methods comparison on an existing, consented dataset rather than an application, and we make no claim that any result benefits Deaf people.

Data. PopSign participants consented to inclusion in a public database, and the dataset card lists identification and appearance-based uses as out of scope; we keep face landmarks local, release only hand and pose landmarks, and never attempt to identify signers. The dataset card also discourages continuous-signing and translation claims, which we avoid.

ASL Citizen. Its licence permits non-commercial research and forbids distributing the data or modifications, requires use consistent with participants' consent, and requires destroying personal data when the research ends. We fetched only the videos needed, deleted them after extracting hand and pose landmarks, extracted no face landmarks, release only aggregate results, and will delete the local landmarks when the work is complete. The dataset's recommended-use page discourages continuous-signing and interpreter-replacement uses and any attempt to identify participants; we do none of these.

Risks. A cheap recogniser could be presented as a substitute for interpreters or language access. High accuracy on 250 isolated signs does not support that, and we discourage it. Community perspectives on automatic sign language translation (Tran et al., 2023) and dataset governance (Bragg et al., 2021) should be consulted before any deployment.

Community feedback. At the time of this version, no Deaf community member or dataset creator has reviewed the framing of this paper. On 13 October 2026 the author sent a one-page plain-language summary to the PopSign team and to a

Deaf researcher working on the ethics of sign language technology, asking whether anything is misleading, missing or objectionable. No reply had been received when this version was written; a later version will report what they said and what changed as a result, or that no reply was received.

9 Reproducibility

Code, configuration, pinned requirements, per-query results and generated tables are in the accompanying repository, archived as release v0.1.0 at doi:10.5281/zenodo.23351799 (`make quick` reruns everything; `make report` regenerates every table, figure and number in this paper from the stored results). Landmark extraction versions and timings are logged, and every result file has a provenance file with the commit and library versions. The pre-registration is `PREREG.md`; deviations from it are listed there. Derived hand and pose landmarks from PopSign are not part of the v0.1.0 archive; if released later it will be under CC BY 4.0 with attribution to Georgia Tech and the Deaf Professional Arts Network. Videos and face landmarks are not released. ASL Citizen landmarks and per-query results are not released (licence); its aggregate tables, figures and macros are generated by the same `make report` command from committed aggregates.

Use of AI tools. An AI coding assistant (Claude, Anthropic) helped write code, tests and a first draft of the text and checked references. The author specified the design, reviewed and ran the code, checked every number against the generated results files and every citation against its publisher or arXiv record, and takes responsibility for the content.

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